

5. Two magnets P and Q, having masses M_P and M_Q respectively, exert forces on each other and have no other external forces acting on them. The force acting on P is F , which gives P an acceleration a .

Which of the following pairs is correct?

	Magnitude of force on Q	Magnitude of acceleration of Q
A	$\frac{M_Q}{M_P} F$	a
B	$\frac{M_P}{M_Q} F$	a
C	F	$\frac{M_Q}{M_P} a$
D	F	$\frac{M_P}{M_Q} a$

Answer: D

By Newton's 3rd Law, force on P = force on Q

$$F = M_P a \quad \dots\dots (1)$$

$$F = M_Q a_Q \quad \dots\dots (2)$$